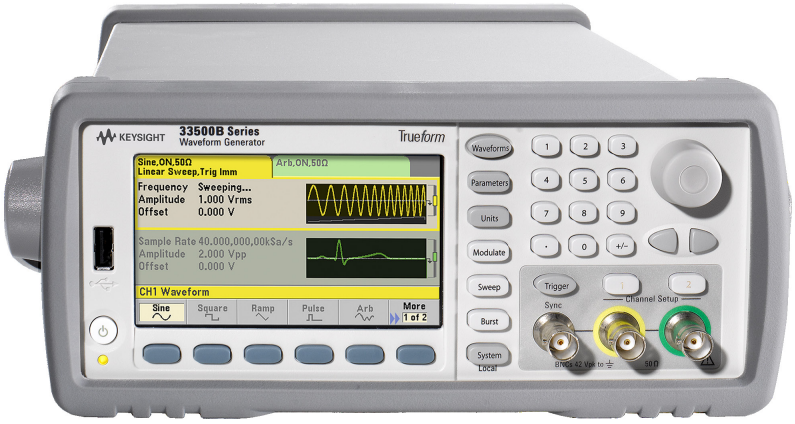
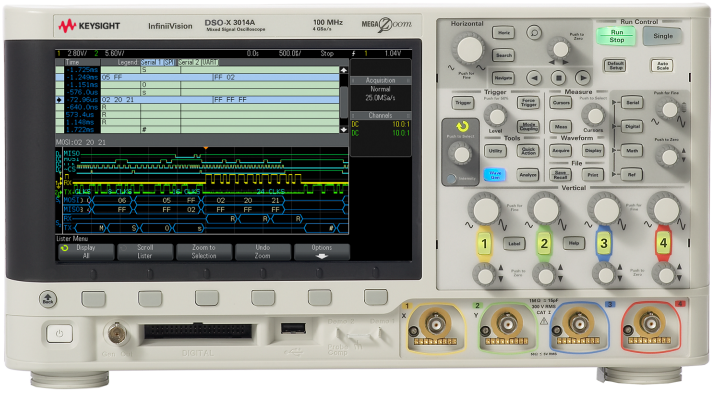
LABORATORY WORKBOOK / 60-75 MINUTES

Phase, interference & quantum

A visual introduction using a waveform generator, an oscilloscope and two BNC cables.



Keysight 33500B Series



Keysight DSO-X 3014A

Create two synchronized waves. Change their relative phase. Measure their time separation. Add them on screen to see reinforcement and cancellation.

WHAT THIS WORKSHOP DEMONSTRATES

You will measure classical electrical waves and use their mathematics to understand quantum interference. A real qubit is not connected: quantum states and probabilities are explained as a separate model.

01 / CONCEPTS BEFORE CONTROLS

Four quantities to recognize

Start here before adjusting the instruments. A cycle is one complete repetition.

Quantity	Meaning	Where to look
Amplitude	Wave height; Vpp is maximum voltage minus minimum voltage.	Vertical direction
Frequency f	Number of complete cycles per second. 1 kHz = 1,000 Hz.	Frequency measurement
Period T	Time for one complete cycle, such as peak to next peak.	Horizontal direction
Relative phase φ	Position of CH2 within a cycle compared with CH1.	Two traces at the same frequency

FREQUENCY AND TIME ARE RECIPROCALS

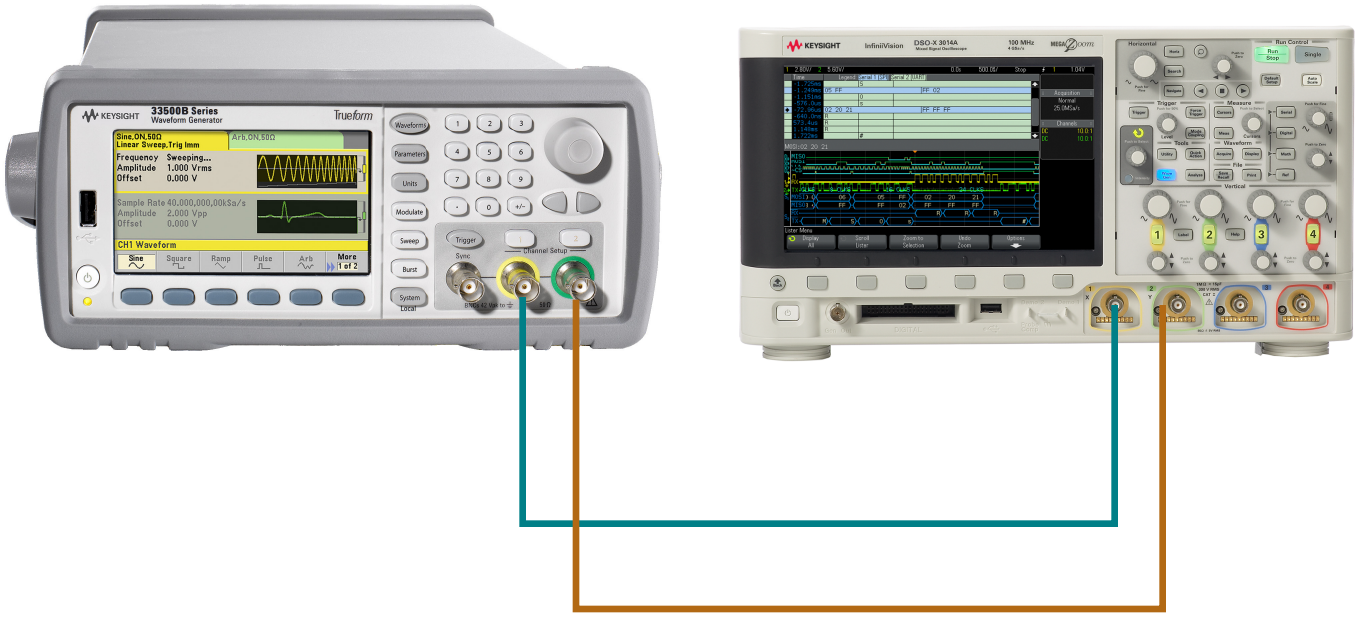
T (seconds) = 1 / f (Hz)
T (milliseconds) = 1 / f (kHz)
T (microseconds) = 1,000 / f (kHz)

Frequency	Period	At 200 µs/div
500 Hz / 0.5 kHz	2 ms / 2,000 µs	10 divisions per cycle
1 kHz	1 ms / 1,000 µs	5 divisions per cycle
2 kHz	0.5 ms / 500 µs	2.5 divisions per cycle

02 / WIRE THE TWO CHANNELS

Two cables. One fixed reference.

Turn both generator outputs OFF before connecting. No BNC tee is needed.



Cable A: generator CH1 → scope CH1

Cable B: generator CH2 → scope CH2

Generator CH1 is the yellow-ring output; CH2 is the green-ring output. On the scope, use the analog inputs numbered 1 and 2. Follow the channel NUMBERS: colors differ between instruments.

IMPORTANT CONNECTION RULE

Do not connect the generator outputs directly together. The addition happens digitally inside the oscilloscope Math function. Leave Sync, external trigger and rear 10 MHz connectors unused for this single-generator experiment.

Equipment: one generator with two working output channels; one scope with two analog inputs; two BNC-to-BNC cables. Keep protective-earth connections intact.

03 / GENERATOR SETUP

Build two matching sine waves.

Use the Channel Setup keys above the BNC outputs, not the keypad digits.

1 Set the load first - on BOTH channels

Press Channel Setup 1. Keep Output OFF. Find Output Load / Load and choose High Z. Use More if needed. Repeat for Channel Setup 2. Set the load before entering amplitude. [1]

2 Enter the same waveform on both channels

Select each channel in turn. Press Waveforms → Sine; then Parameters. Use the screen softkeys, keypad and unit choices to enter the settings below.

Setting	CH1 - reference	CH2 - test signal
Waveform / frequency	Sine / 1.000 kHz	Sine / 1.000 kHz
Amplitude / offset	1.00 Vpp / 0.00 V	1.00 Vpp / 0.00 V
Starting phase	0°	0°
Modulate / Sweep / Burst	OFF / OFF / OFF	OFF / OFF / OFF

3 Synchronize, then enable both outputs

With both channels at the same frequency, use Parameters → Phase → Sync Internal. This aligns their internal phase references. Confirm both phase values are 0°. In each Channel Setup menu, set Output ON. [1]

WHAT “OUTPUT ON” MEANS

It enables the physical BNC signal. A waveform preview alone does not confirm output. “Keep output ON” is an instruction, not a menu label. Softkey names can vary with firmware.

04 / OSCILLOSCOPE SETUP

Make the comparison stable.

Trigger on CH1 throughout. Change only CH2 phase during the phase experiment.

1 Enable channels 1 and 2

Press each numbered channel key. For BOTH inputs select 1 MΩ impedance, DC coupling and 1:1 probe attenuation for a direct BNC connection. Do not leave a 10:1 probe factor selected.

2 Set scales and acquisition

Run acquisition; AutoScale can help find the signals initially. Then set BOTH channels to 200 mV/div and horizontal scale to 200 µs/div. Keep these fixed during comparisons.

3 Use CH1 as the trigger reference

Select an Edge trigger, Source 1, Rising slope and Level 0 V. Set both traces to the same 0 V screen position so matching waves overlap. Use vertical position, not DC offset, for display placement.

4 Add measurements

Press Meas. Choose Source 1 and Type Frequency, then Add Measurement. Repeat for Period and Peak-to-Peak, and check CH2 as well. Expect about 1 kHz, 1 ms and 1 Vpp per channel. [2]

YOUR KEYSIGHT SHOWS SCALES AT THE TOP

200 mV/div means one large vertical square = 0.2 V.
200 µs/div means one large horizontal square = 200 µs.
A 1 Vpp wave spans 5 vertical squares; one cycle spans 5 horizontal squares. Axis numbers in the teaching plots are explanatory labels, not a scope display setting.

05 / PHASE EXPERIMENT

Turn phase into a time shift.

Keep CH1 at 0°. Change CH2 to 0°, 90°, 180°, 270° and 360°.

1 Change CH2, not the reference

On the generator select Channel Setup 2, then Parameters → Phase. Enter 90°. Leave CH1 phase, both frequencies, amplitudes and offsets unchanged. Repeat at each phase value.

2 Measure matching points with time cursors

On the scope press Cursors. Select manual time / X cursors; place X1 at a CH1 rising zero crossing and X2 at the corresponding CH2 rising zero crossing. Read |ΔX|. If cursor source is selectable, assign the appropriate trace. [2]

THE CONVERSION

Phase magnitude (degrees) = 360 × |Δt| / T
Time displacement magnitude = (phase magnitude / 360) × T
Example at 1 kHz: T = 1,000 µs. A quarter-cycle shift is 250 µs = 90°.

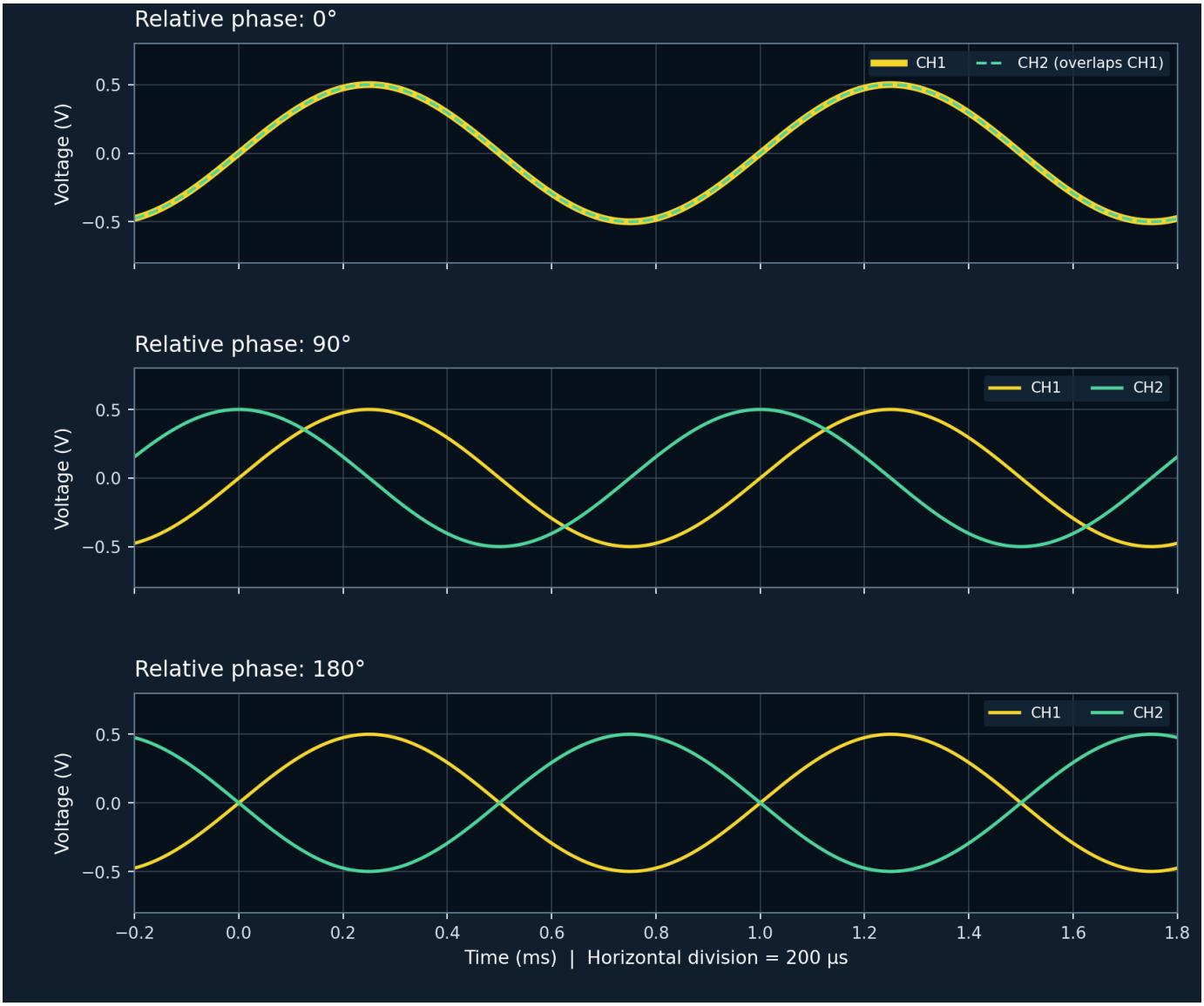
CH2 phase setting	Cycle fraction	Shift at 1 kHz
0°	0	0 µs
90°	1/4 cycle	250 µs
180°	1/2 cycle	500 µs
270°	3/4 cycle	750 µs*
360°	1 full cycle	1,000 µs; overlaps 0°

*Periodic waves repeat: 270° can also be read as a 90° shift in the opposite direction. A 360° shift looks identical to 0°. Check which wave leads; a signed phase reading may use the opposite source order.

06 / EXPECTED PHASE TRACES

Same height. Different alignment.

Idealized traces at 1 kHz and 1 Vpp per channel. Yellow: CH1. Green: CH2.



WHAT TO LOOK FOR

0°: peaks and valleys coincide. 90°: quarter-cycle separation. 180°: a CH1 peak aligns with a CH2 valley. These plots use CH2 = A sin(2πft + φ); positive φ leads CH1. Hardware sign conventions may differ.

07 / ADD THE TWO WAVES

Reveal reinforcement & cancellation.

The scope computes CH1 + CH2 point by point. No physical combiner is required.

1 Keep both original traces active

Confirm both sources are 1 kHz, 1.00 Vpp, 0 V offset, synchronized. Keep CH1 as the trigger. Restore CH2 phase to 0° before enabling the sum.

2 Set up Math addition

Press Math. Select the displayed function f(t), if shown. Choose Operator + / Add, Source 1 = channel 1 and Source 2 = channel 2. Set the Math scale to 500 mV/div and center its zero level. [2]

3 Compare three phases

Change only CH2 phase to 0°, 90° and 180°. Measure the Math trace peak-to-peak voltage using Meas with Math as source, or use voltage cursors. Keep its scale fixed.

Relative phase	Expected sum Vpp	Observation
0°	2.00 Vpp	Maximum reinforcement
90°	About 1.414 Vpp	Partial reinforcement
180°	Ideally 0 Vpp	Cancellation; small residual likely

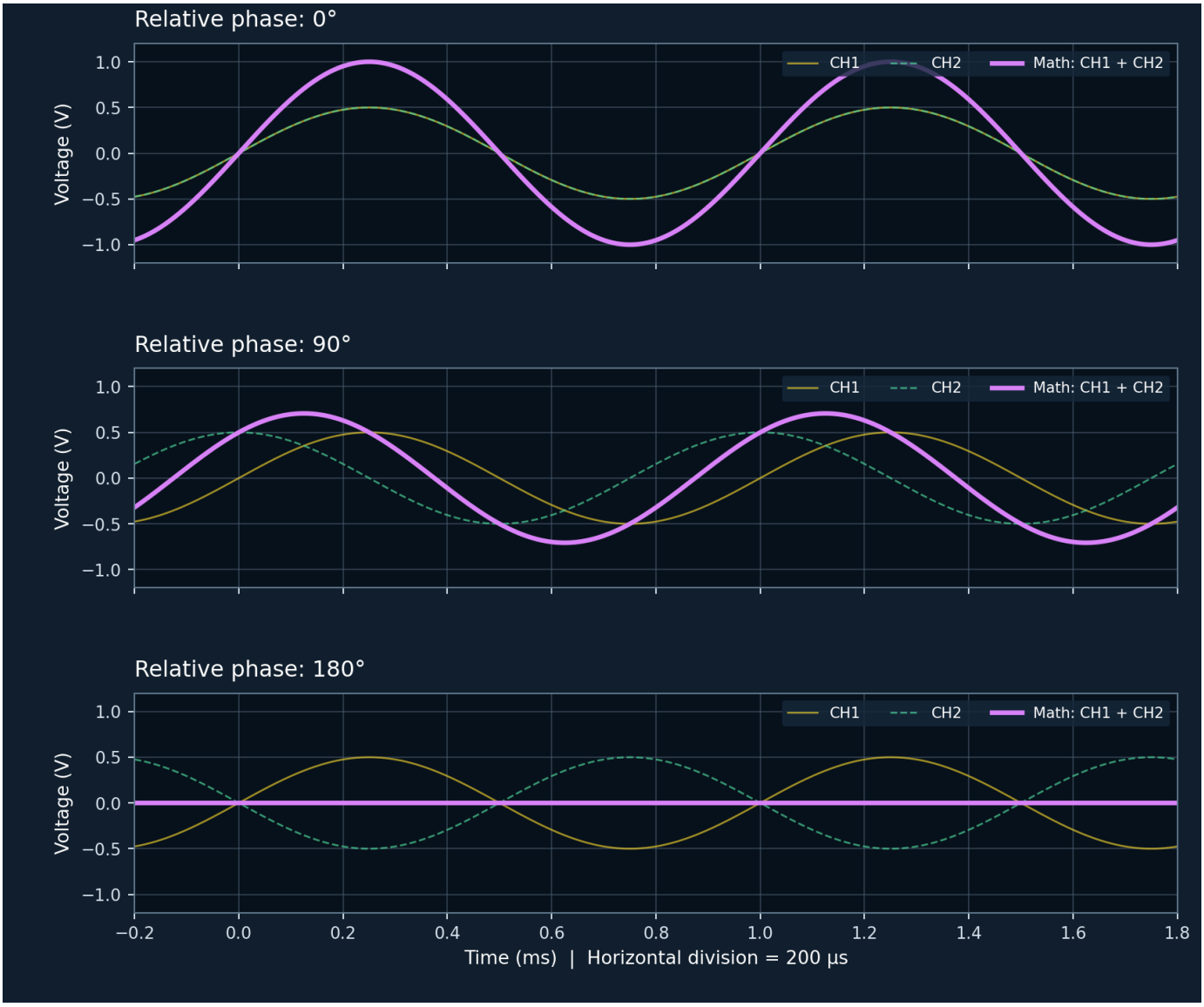
WHY THE SUM MAY NOT REACH ZERO

Perfect cancellation requires equal amplitudes, equal frequencies, matched offsets and 180° relative phase. Noise, gain mismatch and timing error leave a residual. The individual channel voltages remain present.

08 / EXPECTED SUM TRACES

The result depends on phase.

Magenta is the computed sum. The two input amplitudes stay fixed at 1 Vpp each.



MATHEMATICAL CHECK

For two equal sine waves of peak amplitude A:
Sum peak amplitude = 2A × |cos(φ/2)|.
Here A = 0.50 V, so sum Vpp = 2.00 × |cos(φ/2)| volts.

09 / THE QUANTUM ANALOGY

Phase can change an outcome.

A qubit is described by amplitudes. Their squared magnitudes give measurement probabilities. [3]

What we observe here	Corresponding quantum concept
Two classical voltages add.	Superposition: quantum amplitudes combine linearly.
A stable relative phase gives a stable sum.	Coherence preserves meaningful phase relationships.
0° reinforces; 180° cancels the sum.	Interference can increase or suppress an outcome amplitude.
A second channel exposes relative phase.	A recombining operation can reveal relative phase in outcome probabilities.

A precise one-qubit example (theory, not measured here)
Prepare $|0\rangle$ → apply H → apply P(φ) → apply H → measure.
H is the Hadamard gate: it prepares and recombines amplitudes.
P(φ) leaves the $|0\rangle$ amplitude unchanged and gives $|1\rangle$ a relative phase φ.
Before the final H: $(|0\rangle + e^{i\varphi}|1\rangle) / \sqrt{2}$.
After the final H: $P(0) = \cos^2(\varphi/2)$, $P(1) = \sin^2(\varphi/2)$.

Relative phase	Quantum P(0)	Quantum P(1)
0°	100%	0%
90°	50%	50%
180°	0%	100%

The circuit probabilities are theoretical. Direct measurement BEFORE the final H gives 50% / 50% for all these phases. Phase becomes visible through recombination. [3, 4]

10 / LINK THE MATHEMATICS CAREFULLY

A shared pattern, different systems

The classical sum helps visualize the phase dependence of the quantum example.

NORMALIZE, THEN SQUARE - AN OPTIONAL CALCULATION

Measure sum Vpp at phase φ and divide by its 0° value:
$r = Vpp(\text{sum at } \varphi) / Vpp(\text{sum at } 0°)$.
For ideal equal waves: $r = |\cos(\varphi/2)|$, so $r^2 = \cos^2(\varphi/2)$.
This matches the theoretical P(0) curve on page 10.

Worked example at 90°
Measured-model sum ≈ 1.414 Vpp; maximum sum = 2.000 Vpp.
$r \approx 1.414 / 2.000 = 0.707$.
$r^2 \approx 0.500$, matching the theoretical 50% result.

Optional complementary output
Change the Math operator to subtraction, CH1 - CH2. Its normalized squared amplitude follows $\sin^2(\varphi/2)$. At 180° the sum vanishes while the difference is largest. This parallels the complementary probabilities of the circuit.

DO NOT CONFUSE THE ANALOGY WITH A QUANTUM MEASUREMENT

The scope records deterministic voltage, not individual 0/1 quantum outcomes. Two generator channels are not two qubits. Cancellation of a Math trace does not destroy energy in the separate outputs. This setup does not demonstrate entanglement, collapse or quantum randomness.

Also distinguish: classical phase relative to an external timing reference is not the same as an unobservable global phase multiplying an entire quantum state. Here we use a comparison between signals to teach relative phase.

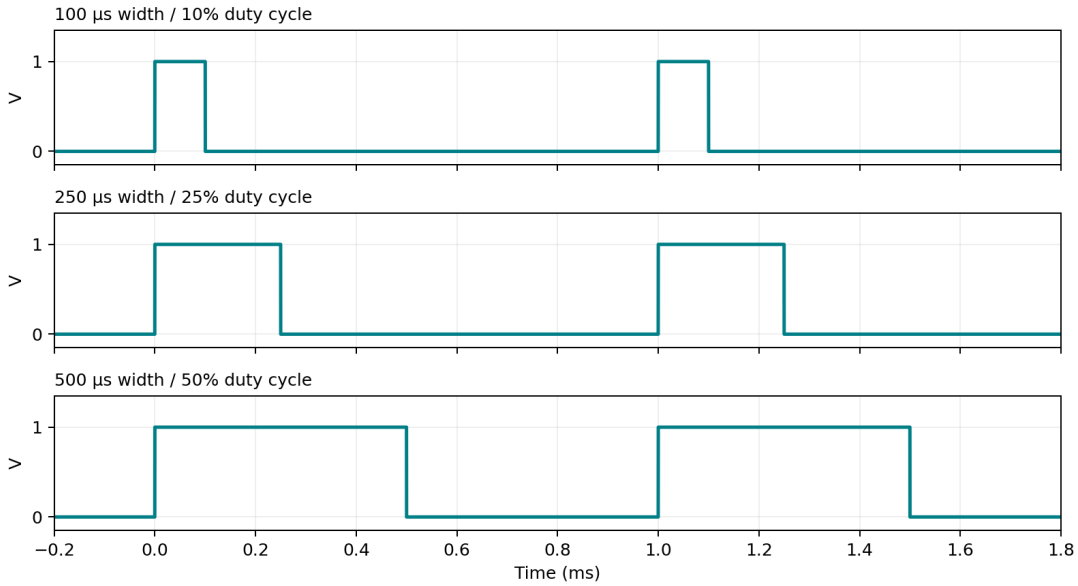
11 / ONE-CABLE REFRESHER

Amplitude, frequency & pulse width

These checks use CH1 alone. Disable CH2 output and hide its trace when returning to them.

Change on generator	Hold fixed	Expected observation
Sine: 2 → 1 Vpp	1 kHz, offset 0 V; 500 mV/div	Height halves from 4 to 2 divisions.
Sine: 1 → 2 kHz	2 Vpp, offset 0 V; 200 µs/div	Period halves: 1 ms → 0.5 ms.
Pulse: width 100 → 250 → 500 µs	1 kHz; 1 Vpp; offset +0.5 V	High level remains +1 V; low level 0 V.

Pulse setup: Waveforms → Pulse; Parameters → Width. Keep Burst OFF and CH1 Output ON. Scope: 200 mV/div, 200 µs/div; CH1 rising-edge trigger at +0.5 V. Use valid short edge times compared with the pulse



At 1 kHz, T = 1 ms. Duty cycle = pulse width / T × 100%. These widths give 10%, 25% and 50%. Duty cycle is a time fraction, not a quantum probability.

12 / STUDENT WORKSHEET

Predict. Measure. Explain.

Group: _____ Date: _____

First record: frequency _____ kHz; period _____ µs.
CH1 _____ Vpp; CH2 _____ Vpp; horizontal scale _____ µs/div.
Trigger source _____; both generator load settings _____.

CH2 phase	Measured |Δt|	Phase from 360|Δt|/T	Sum Vpp	r² (optional)
0°	_____	_____	_____	_____
90°	_____	_____	_____	_____
180°	_____	_____	_____	_____
270°	_____	_____	_____	_____
360°	_____	_____	_____	_____

1. Why did phase seem unchanged with one self-triggered trace?

2. What conditions are needed for cancellation at 180°?

3. Which part is a measured classical result, and which is a quantum model?

13 / INSTRUCTOR NOTES & REFERENCES

Close with the correct explanation.

Answer key, practical fixes and source notes.

Expected results at 1 kHz: 0° / 0 µs / 2 Vpp; 90° / 250 µs / 1.414 Vpp; 180° / 500 µs / near 0 Vpp. At 270°, use 750 µs or a 250 µs shift in the opposite direction; sum ≈ 1.414 Vpp. At 360°, traces overlap 0° and sum ≈ 2 Vpp.

Worksheet answers: 1. Self-triggering realigns each sweep to the same point of one wave. 2. Equal frequencies and amplitudes, matched offsets and 180° relative phase. 3. Voltages, periods and time shifts are measured; quantum probabilities belong to the separate theoretical circuit.

Quick troubleshooting
No trace: confirm correct output/input numbers, Output ON, channel enabled and Run active.
Wrong voltage: check High Z load, 1 MΩ scope input and 1:1 probe factors; re-enter amplitude after changing load.
CH2 drifts: verify identical frequencies; use Parameters → Phase → Sync Internal again.
Phase changes both traces: keep trigger Source 1 and change only generator CH2.
Cancellation is poor: verify Math uses +, check offsets and Vpp, then refine CH2 phase.
Trace disappears at 0°: it may be directly underneath the other trace.

References / further reading
[1] Keysight, Trueform Series Operating and Service Guide, phase synchronization (p. 162), channel setup and load settings. Manufacturer manual hosted by distributor:
33500B / 33600A operating guide
[2] Keysight, InfiniiVision 3000 X-Series User Guide: channels, cursors, measurements and Math (pp. 73-77). Official oscilloscope guide
[3] IBM Quantum Learning, Superposition with Qiskit: amplitudes, phase and interference.
[4] IBM Quantum Learning, Limitations on quantum information: relative versus global phase.
Product photographs: Keysight 33500B datasheet and DSOX3014A product-page image. Photographed screens are illustrative, not the lab settings. All wave plots are calculated teaching figures, not instrument captures. Menu labels may vary by firmware.